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جامعة الأقصى
كلية الحاسبات وتكنولوجيا المعلومات
قسم علم الحاسوب والمعلومات

Operating Systems for Intelligent Systems (ENGS2306)

Assignment 6

Advanced Bash Scripting

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1. Explore Commandes:

1. **Function**: reusable block of code
2. **Calling Function**: Call the function
3. **Parameters**: Pass arguments to function
4. **Local Variable**: Scoped variable inside function
5. **Return Value**: Output from function
6. **Array**: A list of values
7. **Looping Array**: Iterate through array
8. **set -e**: Exit script on error
9. **systemctl**: Check service status

***The complete code of your system_reporter.sh script (copy-paste as text):**

system_reporter.sh

bash

#!/bin/bash

set -e

Define services to check

SERVICES=("ssh" "cron" "ufw")

Function to check service status

```
checkservicestatus() {  
    local service=$1  
    if systemctl is-active --quiet "$service"; then  
        echo -e "[ OK ] $service is active."  
    else  
        echo -e "[FAIL] $service is not active."  
    fi  
}
```

Main script logic

```
echo "System Health Report for $(hostname) as of $(date)"  
echo "-----"
```

```
for service in "${SERVICES[@]}; do  
    checkservicestatus "$service"  
done
```

```
echo "----- End of Report -----"
```

* A screenshot showing the script running successfully :

```
deema@deema-al-maqadma: ~/Desktop
---> System Health Report for deema-al-maqadma as of Thu Aug 14 04:05:48 PM EEST 2025
-----
[ FAIL ] ssh IS NOT ACTIVE.
[ OK ] cron IS ACTIVE.
[ OK ] ufw IS ACTIVE.
-----End of Report -----
deema@deema-al-maqadma:~/Desktop$
```

• A second screenshot showing the :

```
Open system_reporter.sh Save
~/Desktop

1 #!/bin/bash
2 set -e # Stop if an error
3
4 #Define services to check
5 SERVICES=( "ssh" "cron" "ufw" )
6
7 #Function to check service status
8 check_service_status() {
9     local service=$1
10
11     if systemctl is-active --quiet "$service"; then
12         echo -e "[ OK ] $service IS ACTIVE."
13     else
14         echo -e "[ FAIL ] $service IS NOT ACTIVE."
15     fi
16 }
17
18 # Main script logic
19 echo "---> System Health Report for $(hostname) as of $(date)"
20 echo "-----"
21 for service in "${SERVICES[@]}; do
22     check_service_status "$service"
23 done
24
25 echo "-----End of Report -----"
26
27
28
```

9. Report Explanation :

> Using a function like `checkservicestatus()` improves code readability and maintainability. Instead of repeating the same logic inside the loop, the function encapsulates it, making the script cleaner and easier to update if needed.

> Benefits of Using a function:

- **Readability:** The main script logic becomes clean.
- **Reusability:** `check_service()` can be reused in other scripts.
- **Maintenance:** Fix errors in one place.

"واخر دعواهم ان الحمد لله رب العالمين"